WATER RESOURCES, WELLS & PUMPS · NOTE PU01

Water-Lifting Devices — Manual, Animal-Powered and Wind-Powered Lifts

Course topics: CAE1.3 L8, IDE-511 T8, IDE8.16 L6

Classification and choice

Water lifts range from crude home-made devices to efficient factory pumps. By source of power they fall into four groups: **manpower**, **animal power**, **wind power** and **mechanical power** — each group varying widely in design to suit site conditions.

Choose a lift by: the characteristics of the device, the quantity of water, the height of the water table, the type and amount of power available and the farmer's economic condition.

Power source	Relative cost of lifting water
Electric motor pump	Lowest
Engine pump	15–25% higher than electric
Animal power	10–20 times the electric cost
Manual lift	Costliest — 3–4 times the animal cost

- Man and animal power still matter on small holdings, where mechanical skill is scarce, where farmers are poor and for small-scale irrigation in developing countries.
- For large-scale lifting only electric motors and internal-combustion engines are practical.
- **Renewable options** for small-scale lifting at suitable sites: solar photovoltaic, wind, biogas and hydraulic energy.

Manually operated devices

Device	Construction and working	Lift (m)	Discharge	Labour
Swing basket	Basket or shovel-like scoop on four ropes; two people facing each other dip it, swing it up and tip it into a channel	Rarely above 1.0–1.2	3600–5000 L/h	Two persons; labour-intensive and inefficient; for very small lifts and temporary irrigation
Counterpoise lift (dhenkli, picottah)	Long wooden pole pivoted on a support: counterweight (stone, dried mud ball or basket of stones) on the short arm, bucket on the long arm. Pull the long arm down to fill the bucket; the counterweight lifts it; tip it sideways	1–3 (use two or more in series above 3 m)	2000–2300 L/h by one man at 2–3 m lift	One person; for shallow wells, streams and ponds
Don	Counterpoise lift with a boat-shaped wooden or galvanised-iron trough, closed at one end, oscillating about a fixed centre: the closed end dips in, water runs out of the open end	Up to 1.2	—	One person; West Bengal and neighbouring regions
Archimedean screw	Wooden drum with an internal screw turned by a handle on the central spindle; screw 2.0–2.5 m, spindle about 4 m (2.5 cm square), drum about 40 cm diameter, boards held by steel straps every 20–30 cm; inclined at less than 30°; best when the lower end is half submerged	0.5–1.2 (usually 0.5–1.0)	About 16,000 L/h by one man	One person; more efficient and easier than the swing basket
Paddle wheel (chakram)	Paddles on a horizontal shaft push water up a close-fitting concave trough; paddle tips carry a cross-piece about 6 cm wide worked by the feet from a scaffold. Small wheels 1.2 m diameter with about 8 blades; large ones 3.0–3.6 m with up to 24 blades	Rarely above 1.2; usually 0.45–0.6 (45–60 cm)	18,000 L/h with a 2.4 m wheel of 12 blades at 55–60 cm lift, one man	Kerala coast and elsewhere; watering and dewatering paddy

Counterweight of a counterpoise lift

W_1 = weight of bucket, W_2 = weight of water, W_3 = counterweight, Z_1 = movement of the load, Z_2 = movement of the effort, a and b = lever-arm lengths.

$$\text{Velocity ratio} = Z_2/Z_1 = a/b \quad \text{Equilibrium: } (W_1 + W_2)\, Z_1 = W_3\, Z_2 \Rightarrow W_3 = (W_1 + W_2)\, b/a$$

Self-emptying improvement: fix a steel rod with a loop to the bucket; the loop catches a hook at the top and tips the bucket automatically.

Animal-powered devices

Two families: pulley-rope-and-bucket devices, and rotating wheel or drum devices.

Device (local names)	Construction	Lift	Output	Animals and men
Rope-and-bucket lift (mohte, charsa, charus)	Leather or GI bucket of 180–225 L on a stout iron ring and frame, hung from a strong rope over a pulley above the well; bullocks walk down an earthen ramp sloped 5–10°, lifting the bucket, which is tipped by hand into a trough; the rope is then unhitched and the bucket dropped back	Deep wells; can exceed 30 m	≈ 9000 L/h from 15 m (two pairs, three men); depends on depth, animal size and method	1 pair + 2 men, or (commonly) 2 pairs + 3 men — one pair climbs while the other draws
Self-emptying bucket (sundia charas)	Leather spout at the bucket bottom with a second, lighter rope over a roller at the trough; both ropes tied to the yoke. At the top the spout follows its rope over the roller and the bucket empties itself	Up to about 10 m	≈ 8000 L/h from 10 m	1 pair + 1 man; saves one man, but time is lost as bullocks back up the ramp (a toggle that releases the rope speeds it)
Two-bucket lift (circular mohte)	Two buckets of about 70 L with hinged bottom flap valves, raised and lowered alternately by rope, pulley and a central rotating lever, guided by two vertical rods; buckets fill through the flap and are tilted at the top	Up to about 5 m	≈ 14,000 L/h from 4.5 m	1 man + 1 strong or 2 small bullocks; parts of South India
Persian wheel (rahat)	Endless chain of earthen or metal pots (7–14 L) over a large open-spoked drum, driven through a vertical cog wheel meshing with a large horizontal gear turned by animals; a ratchet stops reverse rotation; each pot has a small drain hole	Up to about 10 m; efficiency falls above 7.5 m	≈ 10,000 L/h from 9 m	1 man + 1 pair of bullocks or 1 camel. Drawback: water is lifted above ground before discharge. For lifts ≤ 5 m, two rows of pots on separate drums nearly double output
Chain pump	Endless chain with discs about 25 cm apart passing over a notched wheel and up a pipe of about 10 cm diameter that reaches 0.5–1 m below water; discs (now leather washers backed by iron plates) match the pipe bore; drive like a Persian wheel, often two chains, or a hand crank	Best under 6 m	—	Not popular — heavy wear and low lift
Water wheel (animal-driven)	Paddle wheel in a concave trough, turned through a bullock gear like the Persian wheel	Small lifts	—	Canal areas of Haryana (Rohtak), where fields lie above the distributary

Wind-powered lifts

- Wind is free power wherever it blows steadily above 6.5 km/h. Suitable areas: Rajasthan, Saurashtra (Gujarat), Coimbatore and Kanyakumari districts (Tamil Nadu) and other windy regions.
- **Windmill:** a large vane wheel (rotor) on a tall steel or wooden tower drives a plunger-type reciprocating pump; the plunger rod is fixed eccentrically to the wheel; the pump cylinder, or at least the lower suction pipe, stays under water; the delivery pipe rises to a storage reservoir.
- **Wind needed:** at least 6 to 8 km/h depending on design — assess it before installing. A safety mechanism disengages the rotor in very strong winds. Site it clear of obstructions; output depends on wind velocity and frequency.
- **Use:** mainly lifting water (a little electricity and threshing). Because discharge is intermittent and seasonal, windmills suit only domestic supply and small areas, and need a **storage tank** — water is pumped into it whenever the wind is about 7 km/h or more.
- **Support in India:** the Ministry of New and Renewable Energy (MNRE) runs small-wind and hybrid-system programmes and promotes wind pumps; the **National Institute of Wind Energy (NIWE)**, Chennai, set up in 1998 under MNRE, does R&D on wind energy.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The cheapest way of lifting water is with

(a) an engine pump
(b) an electric motor pump
(c) animal power
(d) manual lifts

Q2. A counterpoise lift is locally called

(a) rahat
(b) dhenkli / picottah
(c) charsa
(d) chakram

Q3. Which manual device lifts about 16,000 L/h at 0.5–1 m lift?

(a) Swing basket
(b) Archimedean screw
(c) Don
(d) Counterpoise

Q4. Persian wheel efficiency falls when lift exceeds about

(a) 2 m
(b) 7.5 m
(c) 20 m
(d) 30 m

Q5. The animal lift suited to wells deeper than 30 m is

(a) Persian wheel
(b) two-bucket lift
(c) rope-and-bucket (mohte)
(d) chain pump

Q6. A windmill pump needs a minimum wind speed of about

(a) 1–2 km/h
(b) 6–8 km/h
(c) 25 km/h
(d) 50 km/h

Answers

Q1 — (b) an electric motor pump
Q2 — (b) dhenkli / picottah
Q3 — (b) Archimedean screw
Q4 — (b) 7.5 m
Q5 — (c) rope-and-bucket (mohte)

Q6 — (b) 6–8 km/h